VSL INTERNATIONAL

Post-Tensioning Systems

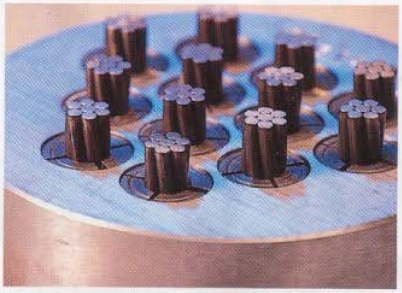


Strand Properties

Strand type		13 mm (0.5")		15 mm (0.6")	
		Euronorm 138-79 or BS 5896: 1980 Super	ASTM A 416-85 Grade 270	Euronorm 138-79 or BS 5896: 1980 Super	ASTM A 416-85 Grade 270
Nominal diameter	(mm)	12.9	12.7	15.7	15.2
Nominal area	(mm^2)	100	98.7	150	140
Nominal mass	(kg/m)	0.785	0.775	1.18	1.10
Yield strength	(MPa)	1580[1]	1670[2]	1500[1]	1670[2]
Tensile strength	(MPa)	1860	1860	1770	1860
Min. breaking load P_N	(kN)	186.0	183.7	265.0	260.7
Young's modulus	(GPa)	circa 195			
Relaxation[3] after 1000 h at 20°C and 0.7 × P_N	(%)	max. 2.5			

1) Measured at 0.1% residual strain (0.1% offset method)
2) Measured at 1.0% extension (1% extension under load method)
3) Valid for relaxation class 2 acc. to Euronorm 138-79/BS 5896:1980, or low relaxation grade acc. to ASTM A 416-85





Tendon Properties

13 mm (0.5") strand				
Tendon unit	Number of strands	Duct diameter[1] internal/ external (mm)	Min. breaking load: Strand type Euronorm 138-79 or BS 5896: 1980, Super (kN)	Min. breaking load: Strand type ASTM A416-85 Grade 270 (kN)
5-1	1	25/30	186	184
5-2	2	40/45	372	367
5-3	3	40/45	558	551
5-4	4	45/50	744	735
5-6[2]	6	50/55	1116	1102
5-7	7	55/60	1302	1286
5-12	12	65/72	2232	2204
5-18[2]	18	80/87	3348	3307
5-19	19	80/87	3534	3490
5-22	22	85/92	4092	4041
5-31	31	100/107	5766	5695
5-37	37	120/127	6882	6797
5-43	43	130/137	7998	7899
5-55	55	140/150	10'230	10'104

15 mm (0.6") strand				
Tendon unit	Number of strands	Duct diameter[1] internal/ external (mm)	Min. breaking load: Strand type Euronorm 138-79 or BS 5896: 1980, Super (kN)	Min. breaking load: Strand type ASTM A416-85 Grade 270 (kN)
6-1	1	30/35	265	261
6-2	2	45/50	530	521
6-3	3	45/50	795	782
6-4	4	50/55	1060	1043
6-6[2]	6	60/67	1590	1564
6-7	7	60/67	1855	1825
6-12	12	80/87	3180	3128
6-18[2]	18	95/102	4770	4693
6-19	19	95/102	5035	4953
6-22	22	110/117	5830	5735
6-31	31	130/137	8215	8082
6-37	37	140/150	9805	9646
6-43	43	150/160	11'395	11'210
6-55	55	170/180	14'575	14'339

1) Standard size for corrugated steel ducts.
For polyethylene ducts PT-PLUS™ see page 6.
2) For intermediate stressing anchorages type Z.

Sheathing and Corrosion Protection

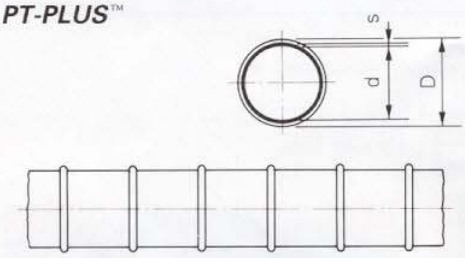
Usually, corrugated steel ducts (sometimes galvanized), with a minimum wall thickness of 0.25 mm are used.

For enhanced corrosion protection and fatigue resistance of the tendons, use of the VSL PT-PLUS™ system with corrugated polyethylene ducts is recommended. The PT-PLUS™ system is particularly suitable for railroad bridges, bridge decks, parking structures and other situations where severe corrosion or high fatigue loading may be expected.

The ducts are manufactured in lengths of 4 to 6 m and are coupled on site.

For diameters of steel ducts, see page 5.

Dimensions of Polyethylene Ducts PT-PLUS™



Strand type 13 mm (0.5")	Strand type 15 mm (0.6")	Dimensions of duct (mm)		
Tendon unit	Tendon unit	d	D	s
5-12	6-7	59	73	2
5-19	6-12	76	91	2.5
5-31	6-19/6-22	100	116	3
Other units on request				

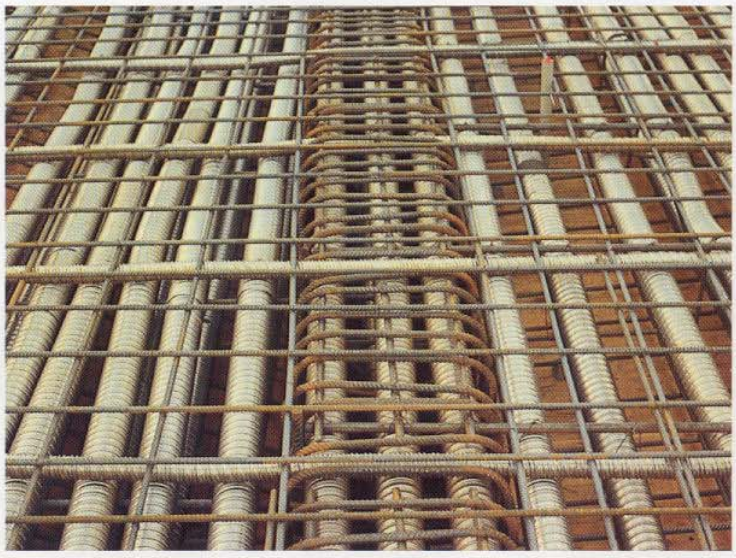


Steel and polyethylene ducts

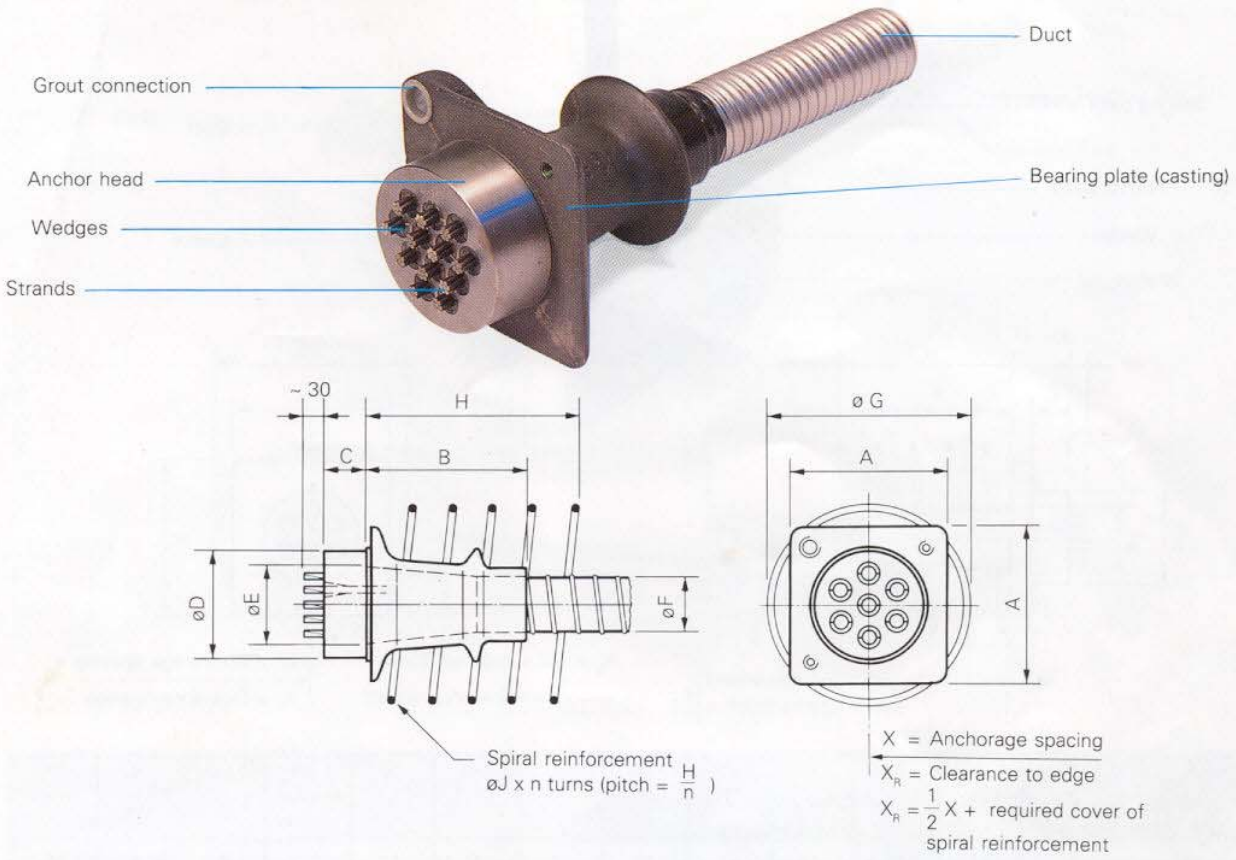


PT-PLUS™ system





Stressing Anchorage VSL Type EC



	Tendon unit	A	B	C	øD	øE	øF[1] internal/ external	øG	H	øJ	n	X[2]
Strand type 13 mm (0.5")	5-3	120	130	50	90	50	40/45	130	150	10	3	155
	5-4	135	125	50	95	55	45/50	160	150	10	3	180
	5-7	165	155	55	110	74	55/60	205	200	12	4	235
	5-12	215	215	60	150	104	65/72	285	250	14	5	305
	5-19	270	285	75	180	135	80/87	365	300	16	6	385
	5-22	290	335	85	190	150	85/92	395	360	18	6	415
	5-31	340	365	95	230	172	100/107	470	400	18	8	490
	5-37	370	360	105	240	188	120/127	510	420	20	7	535
	5-55	430	460	130	290	230	140/150	620	540	22	9	655
Strand type 15 mm (0.6")	6-3	135	125	50	95	55	45/50	160	150	10	3	185
	6-4	150	155	55	110	65	50/55	190	200	12	4	210
	6-7	190	170	60	135	84	60/67	260	250	14	5	280
	6-12	250	245	75	170	118	80/87	345	300	16	6	365
	6-19	310	305	95	200	150	95/102	440	350	18	7	460
	6-22	340	365	100	220	172	110/117	470	400	18	8	495
	6-31	390	350	120	260	192	130/137	560	480	20	8	590
	6-37	430	450	135	280	215	140/150	610	540	22	9	640
	6-55	520	530	160	340	255	170/180	740	630	26	9	780

Dimensions in mm

Subject to modification

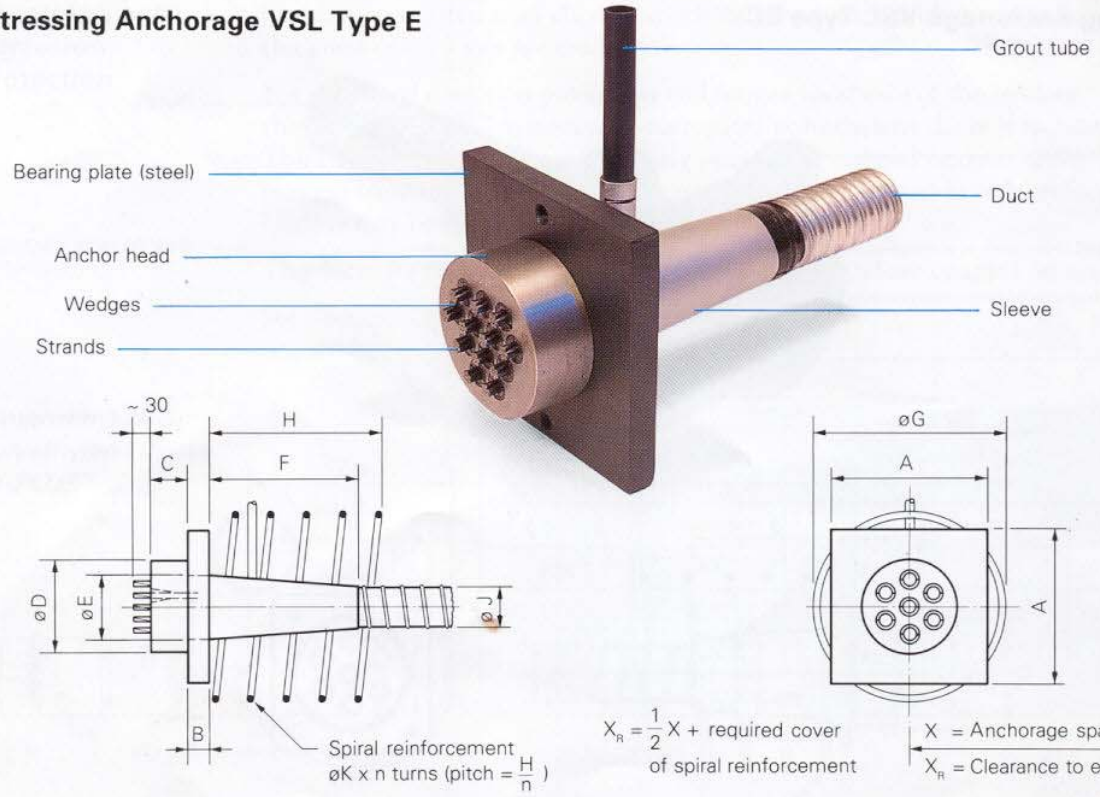
1) Standard diameter for corrugated steel duct. For polyethylene ducts PT-PLUS™ see page 6.
2) Anchorage spacings are in accordance with test requirements of CEB/FIP (Recommendations for acceptance and application of post-tensioning systems; March 1981). Modifications to these values may be possible.

– Dimensions are valid for:
Nominal concrete strength at 28 days: 35 MPa (cube), 28 MPa (cylinder).

Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.
Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).
Dimensions for other concrete strengths on request.

Yield strength of spiral reinforcement: ≥ 420 MPa.
Spirals may be replaced by suitable orthogonal reinforcement.

Stressing Anchorage VSL Type E



	Tendon unit	A	B	C	øD	øE	F	øG	H	øJ[1] internal/ external	øK	n	X[2]
Strand type 13 mm (0.5")	5-1	70	15	45	42	15	70	80	90	25/30	8	2	90
	5-3	115	20	50	90	50	190	130	150	40/45	10	3	155
	5-4	130	20	50	95	55	190	160	150	45/50	10	3	180
	5-7	175	25	55	110	74	190	205	200	55/60	12	4	235
	5-12	230	35	60	150	104	370	285	250	65/72	14	5	305
	5-19	290	40	75	180	135	470	365	300	80/87	16	6	385
	5-22	315	45	85	190	150	480	395	360	85/92	18	6	415
	5-31	370	55	95	230	172	550	470	400	100/107	18	8	490
	5-37	405	60	105	240	188	570	510	420	120/127	20	7	535
	5-43	440	60	110	260	216	680	550	480	130/137	20	8	580
	5-55	500	70	130	290	230	680	620	540	140/150	22	9	655
Strand type 15 mm (0.6")	6-1	75	15	50	53	18	70	80	90	30/35	8	2	105
	6-2	110	15	50	90	50	190	130	150	45/50	10	3	150
	6-3	135	20	50	95	56	190	160	150	45/50	10	3	185
	6-4	160	25	55	110	65	190	190	200	50/55	12	4	210
	6-7	205	35	60	135	84	290	260	250	60/67	14	5	280
	6-12	270	40	75	170	118	460	345	300	80/87	16	6	365
	6-19	340	50	95	200	150	590	440	350	95/102	18	7	460
	6-22	370	55	100	220	172	690	470	400	110/117	18	8	495
	6-31	435	65	120	260	192	690	560	480	130/137	20	8	590
	6-37	480	70	135	280	215	830	610	540	140/150	22	9	640
	6-43	520	75	145	300	246	950	650	640	150/160	26	8	690
	6-55	580	90	160	340	255	950	740	630	170/180	26	9	780

Dimensions in mm

Subject to modification

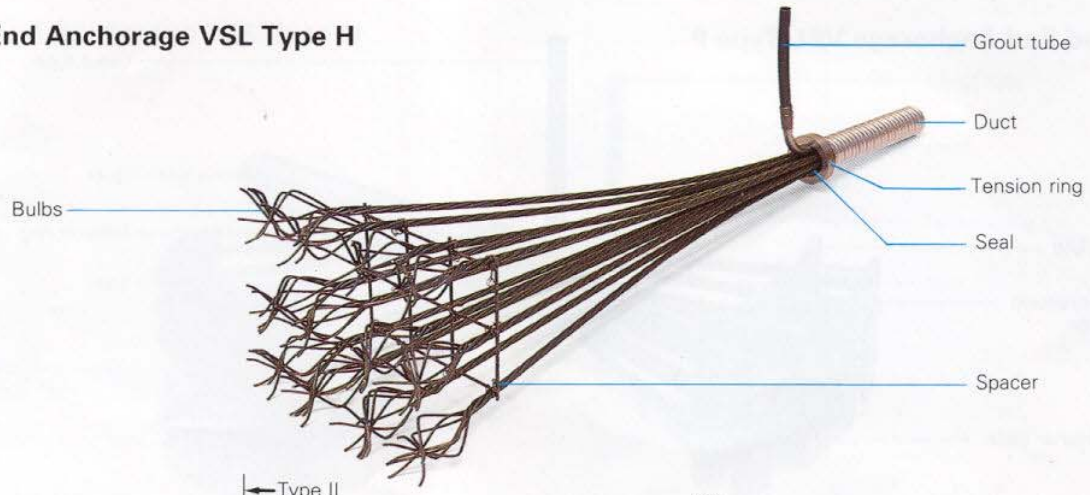
1) Standard diameter for corrugated steel duct. For polyethylene ducts PT-PLUS™ see page 6.

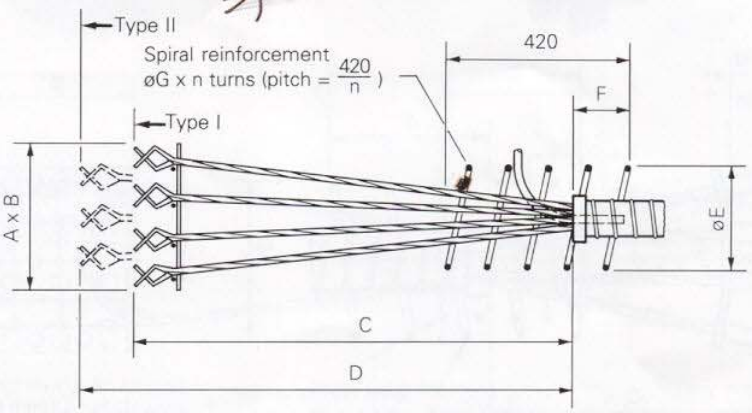
2) Anchorage spacings are in accordance with test requirements of CEB/FIP (Recommendations for acceptance and application of post-tensioning systems; March 1981). Modifications to these values may be possible.

– Dimensions are valid for:
Nominal concrete strength at 28 days: 35 MPa (cube), 28 MPa (cylinder).
Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.
Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).
Dimensions for other concrete strengths on request.

Yield strength of spiral reinforcement: ≥ 420 MPa.
Spirals may be replaced by a suitable orthogonal reinforcement.

Dead End Anchorage VSL Type H





The force in the tendon is transferred by bond of the strands in the concrete and by the bulbs at the end of the strands.

Strand type 13 mm (0.5")										
Tendon unit	Alter-native	Type	A	B	C	D	øE	F	øG	n
5-3	1	I	230	70	930	–	–	–	–	–
5-4	1	I	310	70	930	–	–	–	–	–
	2	I	150	170	930	–	–	–	–	–
5-7	1	II	370	70	1130	1280	180	155	12	7
	2	II	170	190	1130	1280	180	155	12	7
5-12	1	II	350	190	1130	1280	200	155	14	7
	2	I	310	270	1130	–	200	155	14	7
5-19	1	II	470	190	1130	1280	230	155	14	7
	2	II	310	390	1130	1280	230	155	14	7
5-22	1	II	570	190	1130	1280	300	155	16	7
	2	II	390	390	1130	1280	300	155	16	7
5-31	1	II	670	310	1330	1480	350	155	16	7
	2	II	470	430	1330	1480	350	155	16	7
5-37	1	II	770	310	1530	1680	350	165	18	7
	2	II	470	550	1530	1680	350	165	18	7
5-43	1	II	870	350	1530	1680	400	165	18	7
	2	II	670	430	1530	1680	400	165	18	7
	3	II	570	550	1530	1680	400	165	18	7
5-55	1	II	1170	350	1830	1980	400	175	20	7
	2	II	870	430	1830	1980	400	175	20	7
	3	II	570	670	1830	1980	400	175	20	7

Strand type 15 mm (0.6")										
Tendon unit	Alter-native	Type	A	B	C	D	øE	F	øG	n
6-3	1	I	290	90	950	–	–	–	–	–
6-4	1	I	390	90	950	–	–	–	–	–
	2	I	190	210	950	–	–	–	–	–
6-7	1	II	450	90	1150	1300	200	155	14	7
	2	II	210	230	1150	1300	200	155	14	7
6-12	1	II	430	230	1150	1300	230	155	14	7
	2	I	390	330	1150	–	230	155	14	7
6-19	1	II	570	230	1150	1300	300	155	16	7
	2	II	390	470	1150	1300	300	155	16	7
6-22	1	II	690	230	1150	1300	350	155	16	7
	2	II	470	490	1150	1300	350	155	16	7
6-31	1	II	810	260	1550	1700	400	165	18	7
	2	II	570	510	1550	1700	400	165	18	7
6-37	1	II	1050	370	1850	2000	400	175	20	7
	2	II	690	510	1850	2000	400	175	20	7
6-43	1	II	1050	370	1850	2000	400	175	20	7
	2	II	810	510	1850	2000	400	175	20	7
	3	II	690	650	1850	2000	400	175	20	7
6-55	1	II	1410	370	2150	2300	460	185	22	8
	2	II	1050	510	2150	2300	460	185	22	8
	3	II	690	790	2150	2300	460	185	22	8

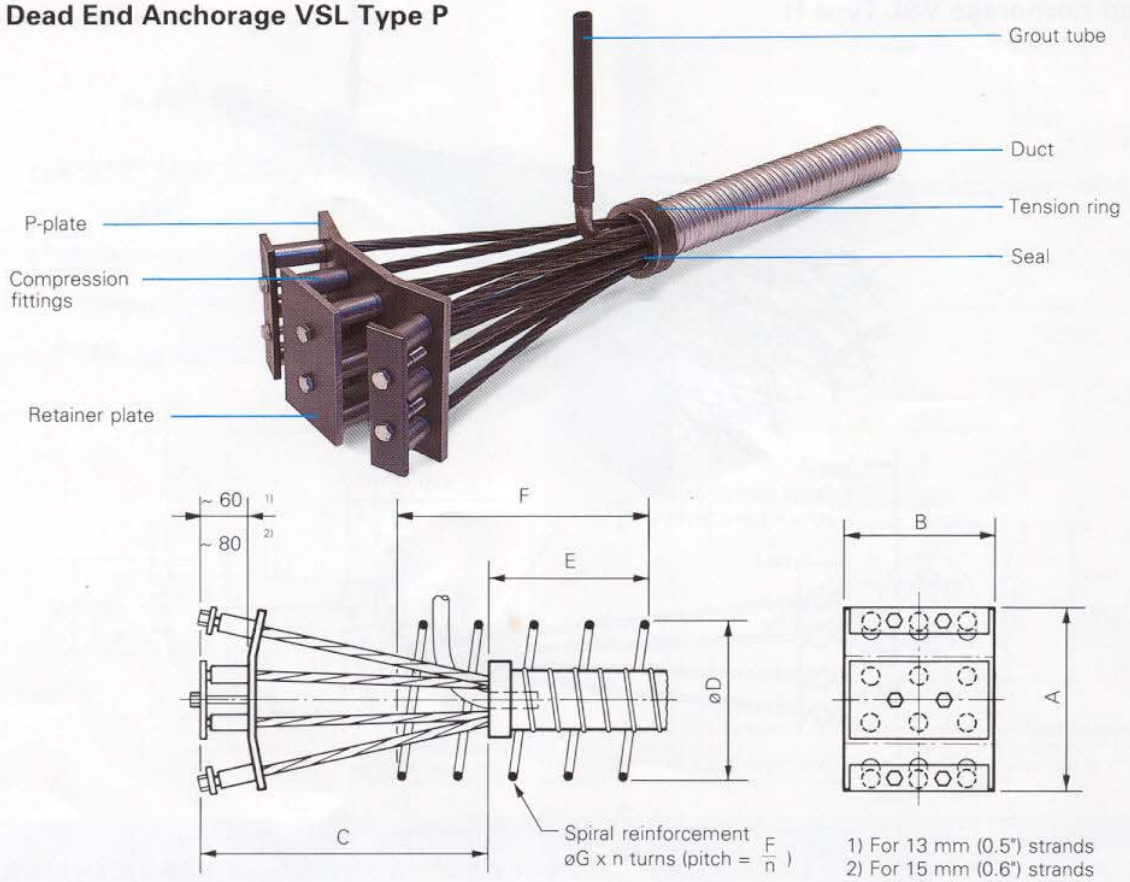
Dimensions in mm

Subject to modification

Dimensions are valid for:
Nominal concrete strength at 28 days: 35 MPa (cube), 28 MPa (cylinder).
Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.

Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).
Dimensions for other concrete strengths on request.
Yield strength of spiral reinforcement: ≥ 420 MPa.
Spirals may be replaced by a suitable orthogonal reinforcement.

Dead End Anchorage VSL Type P



Where the force must be transferred into the concrete exclusively via the bearing plate, polyethylene tubes may be used for sheathing the strands between the end of the duct and the bearing plate.

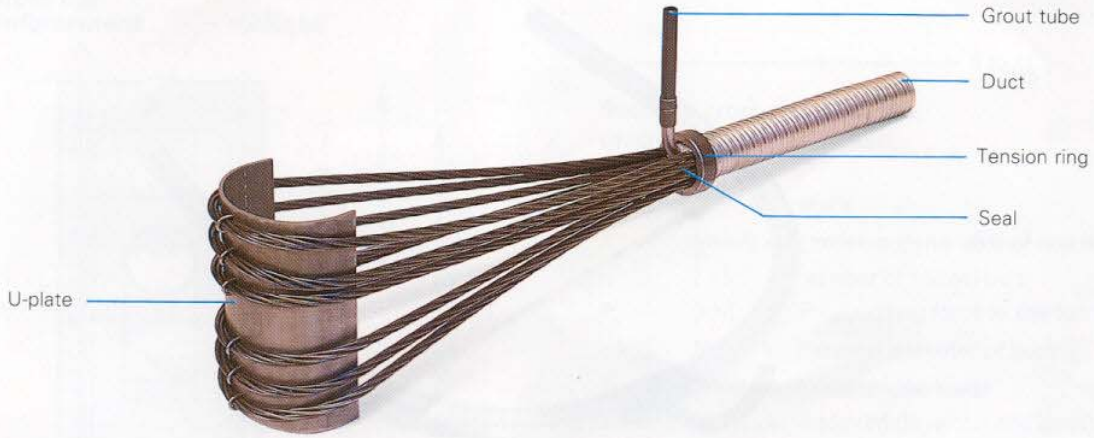
Strand type 13 mm (0.5")									
Tendon unit	Alternative	A	B	C	øD	E	F	øG	n
5-3		190	60	260	130	90	190	8	4
5-4	1	120	120	360	130	90	190	8	4
	2	230	65	360	130	90	190	8	4
5-7		230	110	360	180	100	250	12	4
5-12		230	190	410	200	140	320	14	5
5-19		300	230	510	230	140	320	14	5
5-22		350	230	610	300	140	320	16	5
5-31		500	230	810	300	140	320	16	5
5-37	1	600	230	960	350	170	380	18	6
	2	470	300	810	350	170	380	18	6
5-43	1	690	230	1010	400	170	380	18	6
	2	530	300	860	400	170	380	18	6
5-55	1	880	230	1310	400	170	380	20	6
	2	600	360	910	400	170	380	20	6

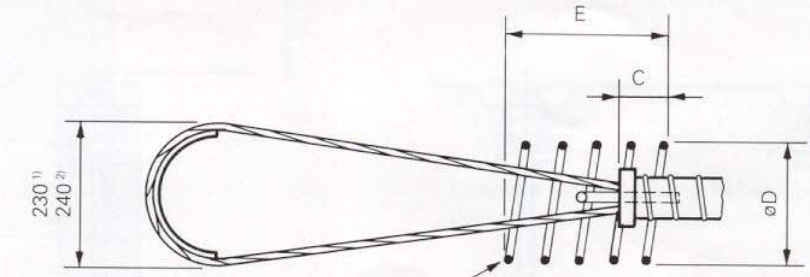
Strand type 15 mm (0.6")									
Tendon unit	Alternative	A	B	C	øD	E	F	øG	n
6-2		180	60	280	130	90	190	8	4
6-3		230	70	330	180	100	250	12	4
6-4	1	260	80	430	180	100	250	12	4
	2	145	145	380	180	100	250	12	4
6-7		270	140	430	200	140	320	14	5
6-12		270	230	430	230	140	320	14	5
6-19		370	270	630	300	140	320	16	5
6-22		420	270	730	300	140	320	16	5
6-31	1	600	270	980	400	170	380	18	6
	2	480	340	780	400	170	380	18	6
6-37	1	720	270	1180	400	170	380	20	6
	2	560	340	980	400	170	380	20	6
6-43	1	800	270	1280	400	200	440	20	7
	2	650	340	1080	400	200	440	20	7
6-55	1	1000	270	1780	460	220	500	22	8
	2	800	340	1280	460	220	500	22	8

Dimensions in mm
Dimensions are valid for:
Nominal concrete strength at 28 days: 35 MPa (cube), 28 MPa (cylinder).
Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.

Subject to modification
Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).
Dimensions for other concrete strengths on request.
Yield strength of spiral reinforcement: ≥ 420 MPa.
Spirals may be replaced by a suitable orthogonal reinforcement.

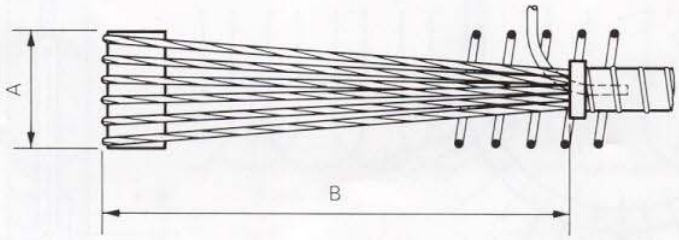
Dead End Anchorage VSL Type U





1) For 13 mm (0.5") strands
2) For 15 mm (0.6") strands

Spiral reinforcement
øF x n turns (pitch = $\frac{E}{n}$)



Strand type 13 mm (0.5")							
Tendon unit	A	B	C	øD	E	øF	n
5-1	15	600	–	–	–	–	–
5-3	70	600	80	130	190	8	4
5-7	160	600	110	190	260	12	5
5-12	280	700	110	220	260	14	5
5-19	400	800	130	260	320	16	5
5-22	440	900	130	260	320	16	5
5-31	640	1300	130	260	320	16	5

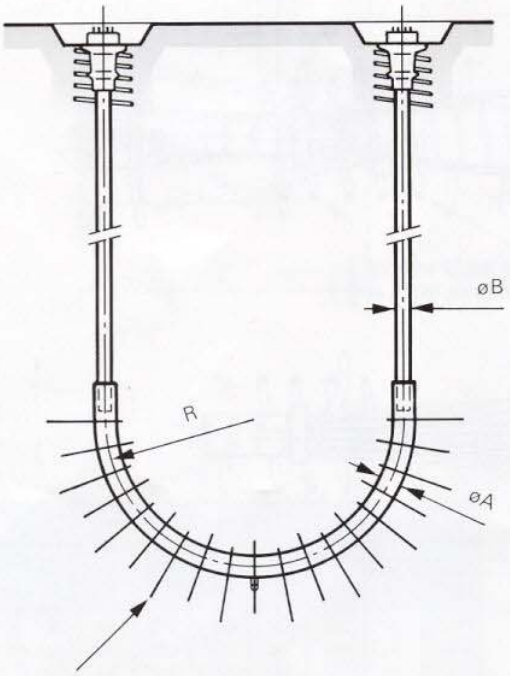
Dimensions in mm
Dimensions are valid for:
Nominal concrete strength at 28 days: 35 MPa (cube), 28 MPa (cylinder).
Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.

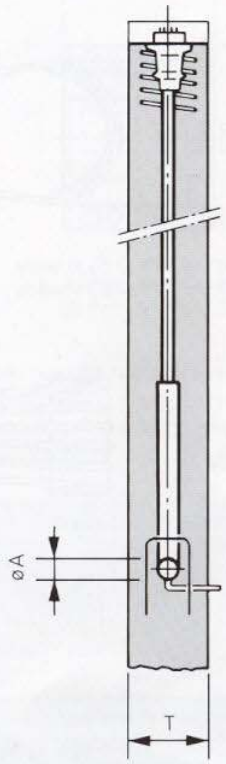
Strand type 15 mm (0.6")							
Tendon unit	A	B	C	øD	E	øF	n
6-1	35	600	–	–	–	–	–
6-2	75	600	80	130	190	8	4
6-3	145	600	110	190	260	12	5
6-4	145	600	110	190	260	12	5
6-7	255	700	110	220	260	14	5
6-12	400	900	130	260	320	16	5
6-19	690	1450	130	260	320	16	5

Subject to modification
Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).
Dimensions for other concrete strengths on request.
Yield strength of spiral reinforcement: ≥ 420 MPa.
Spirals may be replaced by a suitable orthogonal reinforcement.

Dead End Anchorage VSL Type L







Strand type 13 mm (0.5")			
Tendon unit	øA internal/external	øB internal/external	R min.
5-4	50/55	45/50	600
5-7	65/72	55/60	600
5-12	80/87	65/72	900
5-19	90/97	80/87	1100
5-22	95/102	85/92	1200
5-31	110/117	100/107	1400

Strand type 15 mm (0.6")			
Tendon unit	øA internal/external	øB internal/external	R min.
6-2	50/55	45/50	600
6-3	50/55	45/50	600
6-4	55/60	50/55	600
6-7	75/82	60/67	750
6-12	90/97	80/87	1000
6-19	110/117	95/102	1300

Subject to modification

Dimensions in mm
Dimensions are valid for:
Nominal concrete strength at 28 days: 25 MPa (cube), 20 MPa (cylinder).
Maximum prestressing force may be applied when concrete reaches 80 % of its nominal strength.
Max. prestressing force is 75 % of min. tendon breaking load (temporary overstressing to 80 %).